THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — SCIENCE · HARD LEVEL CHAPTER TEST SET 3

Chapter 6: **Materials Around Us** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Why is jute used for making sacks? 1

(a) it is transparent **(b)** it is brittle
(c) it is soluble **(d)** it is strong and cheap

2. A fabric that does not absorb water is best for making: 1

(a) raincoats **(b)** handkerchiefs **(c)** bandages **(d)** towels

3. Why is aluminium foil used to wrap food? 1

(a) it can be rolled into thin sheets and is non-toxic
(b) it is transparent
(c) it dissolves
(d) it is magnetic

4. A material that is rough, hard and absorbs water could be: 1

(a) plastic **(b)** glass **(c)** steel **(d)** a brick

5. A fabric that absorbs sweat is best for: 1

(a) tents **(b)** umbrellas **(c)** raincoats **(d)** innerwear

6. Which property is common to a mirror and a polished steel plate? 1

(a) solubility **(b)** lustre **(c)** floating **(d)** transparency

7. Which pair of properties best suit a good insulating handle? 1

(a) transparent and soluble
(b) shiny and magnetic
(c) good conductor of heat
(d) poor heat conductor and non-conductor of electricity

8. Which of the following is an example of using a property of material wisely? 1

(a) using wood for a stove **(b)** using chalk for tyres
(c) using glass for a greenhouse **(d)** using paper for pipes

9. **Assertion (A):** A steel spoon placed in hot tea soon becomes hot. 1
Reason (R): Metals are good conductors of heat.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** A cooking pan has a plastic or wooden handle. 1
Reason (R): Plastic and wood are poor conductors of heat.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why do we call cork a 'light' material? 2

12. Name the property used to separate iron filings from sand and the property used to separate sand from salt. 2

13. Why can't we make a pot from sugar? 2

14. Why is a paper bag unsuitable for carrying water? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. How would you test whether a cloth is cotton or nylon without using fire? 3

16. Why must the same amount of water and heat be used when comparing rods for conduction? 3

17. Explain why transparent materials may still cast faint shadows. 3

OR

Why is iron used for making a plough while cotton is used for making clothes? 3

18. Explain with examples why lustre alone cannot identify a metal. 3

OR

Why are objects painted? Link to properties of materials. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A teacher in Surat sorted objects into two groups. Group A: iron nail, copper wire, aluminium foil, all hard and shiny. Group B: wooden stick, plastic comb, rubber band, none shiny. Group A objects also conducted electricity in a simple circuit while Group B did not. 5
(i) What is the property of shine in metals called?
(ii) Which group contains metals?
(iii) Name a natural material from the story.
(iv) Why are electric wires made of copper but covered with plastic?

20. A shopkeeper says 'Gold is hard like diamond'. Test the claim with properties. 5

OR

Discuss how reuse and recycling of materials such as paper, glass, metals and plastic help the environment. 5

END OF QUESTION PAPER — Total: 40 Marks